

Course: BTech

Semester: 3

Prerequisite: Fundamentals of Computer Systems

Rationale : This course is an introduction to the theory and practice behind modern computer operating systems. Topics will include what an operating system does (and doesn't) do, system calls and interfaces, processes, concurrent programming, resource scheduling and management, virtual memory, deadlocks, algorithms, programming, and security. The approach of the subject is from both a theoretical perspective as well as a practical one.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	-	-	-	3	20	20	-	60	-	100

SEE - Semester End Examination, T - Theory, P - Practical

Course Content

W - Weightage (%) , T - Teaching hours

Sr.	Topics	W	T
1	OS Fundamentals & Architecture OS concepts, generations, types (batch/time-sharing/distributed/real-time). OS services, system calls (process/file/device). OS structures (layered/monolithic/microkernel/hybrid). Virtualization (Type 1/2 hypervisors: VMware/Xen/KVM/VirtualBox). Cloud OS (AWS EC2, Azure VMs, GCE). Mobile OS (Android: Linux kernel/HAL, iOS: Darwin/Core OS).	8	4
2	Processes, Threads & CPU Scheduling Process fundamentals, states (new/ready/running/waiting/terminated), PCB, context switching. Threads (user-level/kernel-level), multithreading models (many-to-one/one-to-one/many-to-many). Scheduling criteria (CPU utilization, throughput, turnaround/waiting/response time). Algorithms: FCFS, SJF/SRTF, Round Robin, Priority (aging). MLQ, MLFQ. Multicore scheduling (load balancing, processor affinity). Real-time (RM, EDF). Linux CFS.	18	8
3	Process Synchronization & Deadlocks Critical section, race conditions, mutual exclusion. Hardware solutions (test-and-set, compare-and-swap). Peterson's solution. Semaphores (counting/binary, P/V operations). Classical problems (Producer-Consumer, Readers-Writers, Dining Philosophers). Monitors, condition variables. Modern primitives (mutexes, spinlocks). Atomic operations, lock-free algorithms. Deadlock: necessary conditions (mutual exclusion, hold-and-wait, no preemption, circular wait). Prevention, avoidance (Banker's algorithm), detection (wait-for graph), recovery.	18	8
4	Memory Management Techniques Logical vs physical address, MMU. Contiguous allocation (fixed/variable partitioning, fragmentation, compaction). Paging (page table, PTBR, protection: valid-invalid bit, sharing). Segmentation (segment table: base/limit). Segmentation with paging (x86). TLB (associative memory, hit ratio). Memory protection (privilege levels: Ring 0 kernel, Ring 3 user).	16	7
5	Virtual Memory & Performance Virtual memory fundamentals, demand paging, page fault handling. Locality (spatial/temporal). Working set model, dirty bit. Copy-on-write. Page replacement: Optimal, FIFO (Belady's anomaly), Second Chance, Clock, NRU, LRU (approximation). Frame allocation (equal/proportional/priority). Global vs local replacement. Thrashing (working set, PFF). Memory-mapped files. Kernel allocation (buddy system, slab allocator). NUMA basics.	18	8
6	Storage, File Systems & Security I/O hardware (device controllers, DMA). I/O software (interrupt handlers, device drivers). File concepts, access methods (sequential/direct/indexed), directory structures. File allocation (contiguous/linked/indexed: FAT/inode). Free-space (bitmap/linked list). Disk structure, scheduling (FCFS, SSTF, SCAN, C-SCAN, LOOK). SSD (wear leveling, garbage collection), NVMe. RAID (0/1/5/6/10). File systems (ext4, NTFS, FAT32), journaling. OS security: authentication (MFA), access control (ACL, DAC/MAC/RBAC), encryption (TLS/SSL). Containers (Docker, namespaces: PID/network/mount, cgroups).	22	10
Total		100	45

**Reference Books**

1.	Operating System Concepts Essentials (TextBook) By byAviSilberschatz, Peter Galvin,Greg Gagne 9th Edition Wiley Asia Student Edition.
2.	Operating Systems Internals and Design Principles By William Stallings PHI 5th Edition
3.	3. Operating System: A Design-oriented Approach By Charles Crowley Irwin Publishing ,; 1st Edition -
4.	Operating Systems: A Modern Perspective By by Gary J. Nutt, Addison-Wesley 2nd Edition
5.	Design of the Unix Operating Systems By Maurice Bach, Prentice-Hall of India 8th Edition
6.	Understanding the Linux Kernel By Daniel P. Bovet, Marco Cesati, O'Reilly and Associates 3rd Edition

Course Outcomes

At the end of this course Students Will be able to:

1	Distinguish operating system architectures, services, and virtualization techniques.
2	Explain process management, threads, and CPU scheduling techniques.
3	Analyze process synchronization mechanisms and deadlock handling techniques.
4	Explain memory management techniques and address translation mechanisms.
5	Analyze virtual memory management and page replacement techniques.
6	Explain storage management, file systems, and operating system security mechanisms.